

Q1 AI-Based Student Performance Analytics System

AI-Based Student Performance Analytics System ♦ Scenario: Intelligent Education. A university wants to develop an AI-assisted Student Performance Analytics System to monitor student academic performance and identify students who may need additional academic support.

- A. Create a class Student with attributes such as Student ID, Name, Marks in different subjects, Average, and Performance Level.
- B. Implement member functions to input student details, calculate average marks, determine performance level, and display the student report.
- C. Use an array of objects to manage records of multiple students.
- D. Include a static member to maintain the total number of students analyzed.
- E. Apply proper encapsulation using private data members and public member functions.



```

1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Student {
6 private:
7     int studentID;
8     string name;
9     float marks[3];
10    float average;
11
12 public:
13     static int totalStudents;
14
15     void input() {
16         cout << "Enter Student ID: ";
17         cin >> studentID;
18
19         cout << "Enter Name: ";
20         cin >> name;
21
22         cout << "Enter marks in 3 subjects: ";
23         for (int i = 0; i < 3; i++)
24             cin >> marks[i];
25
26         calculateAverage();

```

```

2
101
Gopinath
85 90 80

```

Output

Q2 Digital Healthcare Patient Monitoring System

10 marks

Digital Healthcare Patient Monitoring System ♦ Scenario: Smart Healthcare. A modern hospital wants to implement a Digital Patient Monitoring System that maintains patient information and automatically calculates basic health-risk indicators from recorded medical parameters.

- A. Design a class Patient with attributes such as Patient ID, Name, Age, Heart Rate, Blood Pressure, and Health Status.
- B. Implement member functions to input patient details, calculate/identify the health status based on predefined conditions, and display the patient report.
- C. Store multiple patient records using an array of objects.
- D. Use a static member to track the total number of patients monitored.
- E. Demonstrate data hiding, encapsulation, and structured object-oriented program design.

our Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Patient {
6 private:
7     int patientID;
8     string name;
9     int age;
10    float heartRate;
11    float bloodPressure;
12    string healthStatus;
13
14 public:
15     static int totalPatients;
16
17     void input() {
18         cin >> patientID;
19         cin >> name;
20         cin >> age;
21         cin >> heartRate;
22         cin >> bloodPressure;
23
24         calculateStatus();
25         totalPatients++;
26     }
```

input

```
2
101
Gopinath
20
```

Output



Assessment

Q3 Digital Wallet and FinTech Transaction System

10

Digital Wallet and FinTech Transaction System ♦ Scenario: Cashless Economy. A FinTech company wants to develop a Digital Wallet Management System to maintain customer wallet information and perform secure financial transactions.

- A. Create a class DigitalWallet with attributes such as Wallet ID, Customer Name, Mobile Number, and Balance.
- B. Implement member functions for wallet creation, money deposit, money withdrawal, balance enquiry, and transaction display.
- C. Use an array of objects to manage multiple customer wallets.
- D. Include a static member to maintain the total number of active wallets.
- E. Apply appropriate access specifiers and encapsulation to prevent direct modification of sensitive financial data.

Your Code

Input

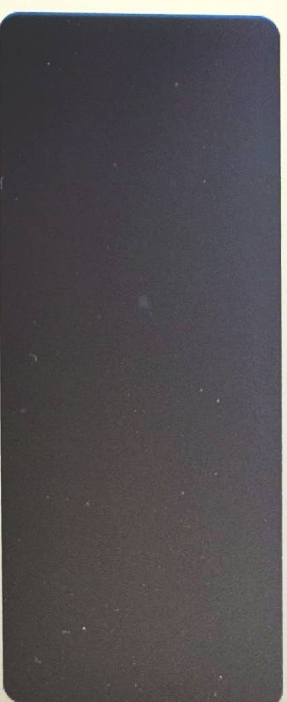
Your Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class DigitalWallet {
6 private:
7     int walletID;
8     string customerName;
9     string mobileNumber;
10    double balance;
11
12 public:
13     static int totalWallets;
14
15     void createWallet() {
16         cin >> walletID;
17         cin >> customerName;
18         cin >> mobileNumber;
19         cin >> balance;
20         totalWallets++;
21     }
22
23     void deposit(double amount) {
```

Input

```
2
101
Gopinath
9876543210
```

Output



Q4 Smart Agriculture Crop Monitoring System

Smart Agriculture Crop Monitoring System ♦ Scenario: IoT-Based Agriculture. An agricultural research centre is developing an IoT-based Smart Agriculture System to monitor crop fields using sensor data and identify whether crops require irrigation.

- A. Design a class CropField with attributes such as Field ID, Crop Name, Soil Moisture, Temperature, Humidity, and Irrigation Status.
- B. Implement member functions to input sensor readings, evaluate soil conditions, determine irrigation requirements, and display field information.
- C. Use an array of objects to maintain information about multiple crop fields.
- D. Include a static member to count the total number of monitored fields.
- E. Ensure proper encapsulation and modular object-oriented design suitable for an IoT-based agricultural application.

Your Code

Input

Your Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class DigitalWallet {
6 private:
7     int walletID;
8     string customerName;
9     string mobileNumber;
10    double balance;
11
12 public:
13     static int totalWallets;
14
15     void createWallet() {
16         cin >> walletID;
17         cin >> customerName;
18         cin >> mobileNumber;
19         cin >> balance;
20         totalWallets++;
21     }
22
23     void deposit(double amount) {
24         balance += amount;
25     }
26 }
```

Input

```
2
101
Gopinath
9876543210
```

Output



Q5 Smart City IoT Energy Management System

10 mark

Smart City IoT Energy Management System ♦ Scenario: Sustainable Smart City. A smart-city research project aims to develop an IoT-based Energy Management System for monitoring electricity consumption in different buildings and identifying buildings with high energy usage.

- A. Create a class Building with attributes such as Building ID, Building Name, Building Type, Energy Consumption, and Energy Status.
- B. Implement member functions to input building details, calculate energy status based on consumption, and display energy usage information.
- C. Use an array of objects to manage multiple buildings in the smart-city system.
- D. Include a static member to maintain the total number of buildings monitored.
- E. Apply encapsulation, data hiding, and structured C++ object-oriented programming principles in the implementation.

Your Code

Input

Our Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Building {
6 private:
7     int buildingID;
8     string buildingName;
9     string buildingType;
10    float energyConsumption;
11    string energyStatus;
12
13 public:
14     static int totalBuildings;
15
16     void input() {
17         cin >> buildingID;
18         cin >> buildingName;
19         cin >> buildingType;
20         cin >> energyConsumption;
21
22         calculateStatus();
23         totalBuildings++;
24     }
25
26     void calculateStatus() {
27         if (energyConsumption <= 100)
```

Input

```
2
101
GreenBuilding
Office
--
```

Output

